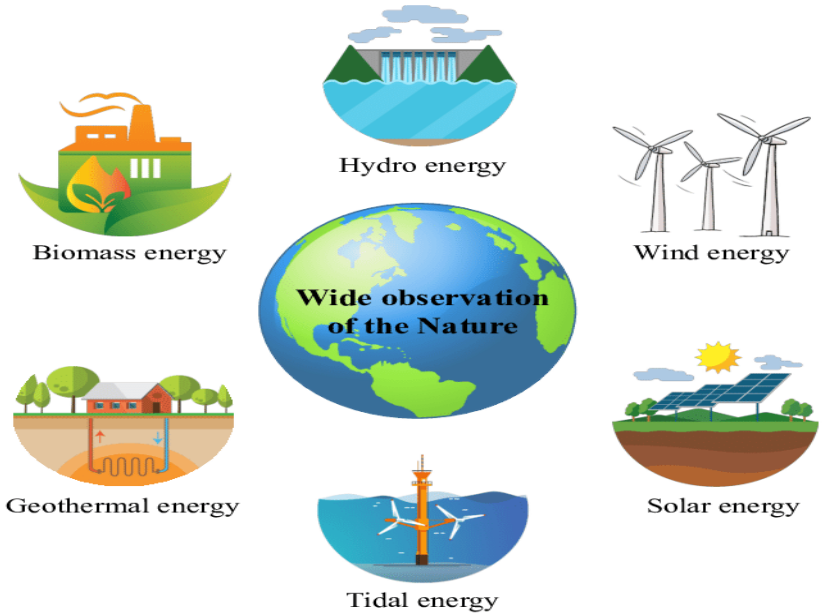
GE22451- Environmental sciences and Sustainability

Unit 3

RENEWABLE SOURCES OF ENERGY



Unit syllabus

- Energy resources: Growing energy needs, Non-renewable resources – types, uses. Energy management and conservation –
- New energy sources, Need of new sources - geo suitability of establishing renewable energy sources, different types of new energy sources.
- Applications of hydrogen energy, ocean energy resources, Tidal energy conversion.
- Concept, origin and power plants of geothermal energy.
- Role of an individual in conservation of energy.

NATURAL RESOURCES session 19

Natural resources are the sources which are useful to man or can be transformed into a useful product.

Natural resources are classified into two types.

1. Renewable resources.
2. Non-renewable resources.

1. Renewable resources

Resources that are capable of being regenerated by ecological processes within a reasonable time period are called renewable resources. They have the potential to renew themselves

Examples - Soil, water, air, wildlife, natural vegetation.

The renewable resources are further sub classified into two types.

i. Continuous resources

These resources are continuously renewed.

Examples - Solar energy, wind, tidal energy.

ii. Extrinsic resources

These resources are prone to breakdown or degradation, yet are available continuously if well managed.

Example – Human skills, institutions, management abilities

2. Non-renewable resources

Resources that are not capable of being regenerated by ecological processes are called non-renewable resources.

Examples - Minerals, coal, oil, natural gas, ground water.

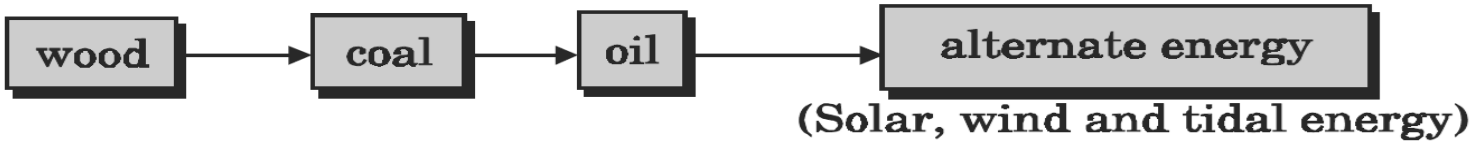
Energy Resources

- Energy is defined as “the capacity to do work”

- Energy is available on the earth in a number of forms.

- All developmental activities in the world are directly or indirectly dependent upon energy. Both production and energy utilization are the indicators of a country’s progress.

- The first form of energy is fire. Wood is the main source of energy, which is later replaced by coal. Coal is now being replaced by oil and gas.



Growing energy needs

- All industrial process like, mining, transport, lighting, heating and cooling in buildings, all require energy.

- With the demands of growing population, the world is facing further energy deficit.

- Our life style is also changing from a simple way of life to a luxurious life style.

- At present 95% of the commercial energy is available only from the fossil fuels like coal, oil and natural gas, and are not going to last for many more years.

Energy Distribution-World Scenario

- Developed countries like U.S.A and Canada constitute only 5% of the world's population, but consume 25% of the available world's energy resources.

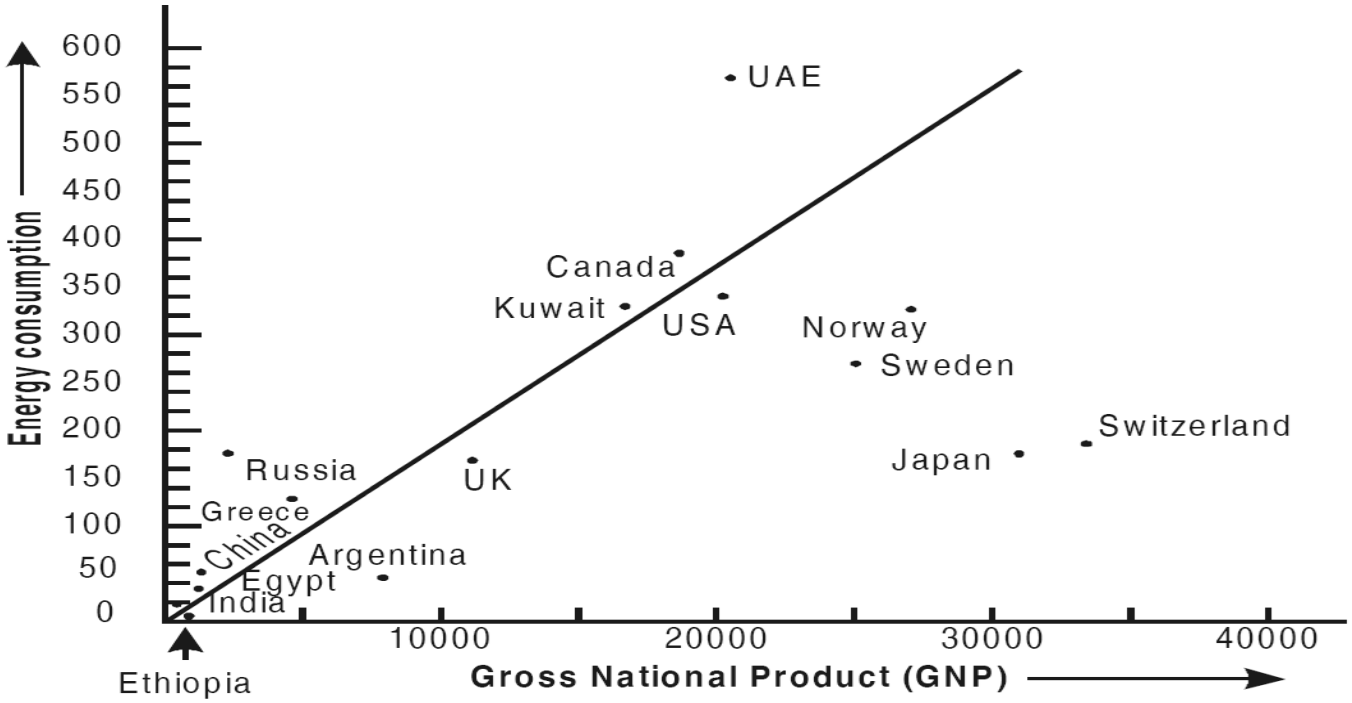
- It has been observed, that in USA and Canada an average person consumes 300 GJ (Giga Joules; equal to 60 barrels of oil) per year.

- But in poor countries like Bhutan, Nepal and Ethiopia, an average person consumes less than 1 GJ per year. So a person in a developed country consumes almost as much energy in a single day as one person consumes in a whole year in a poor country.

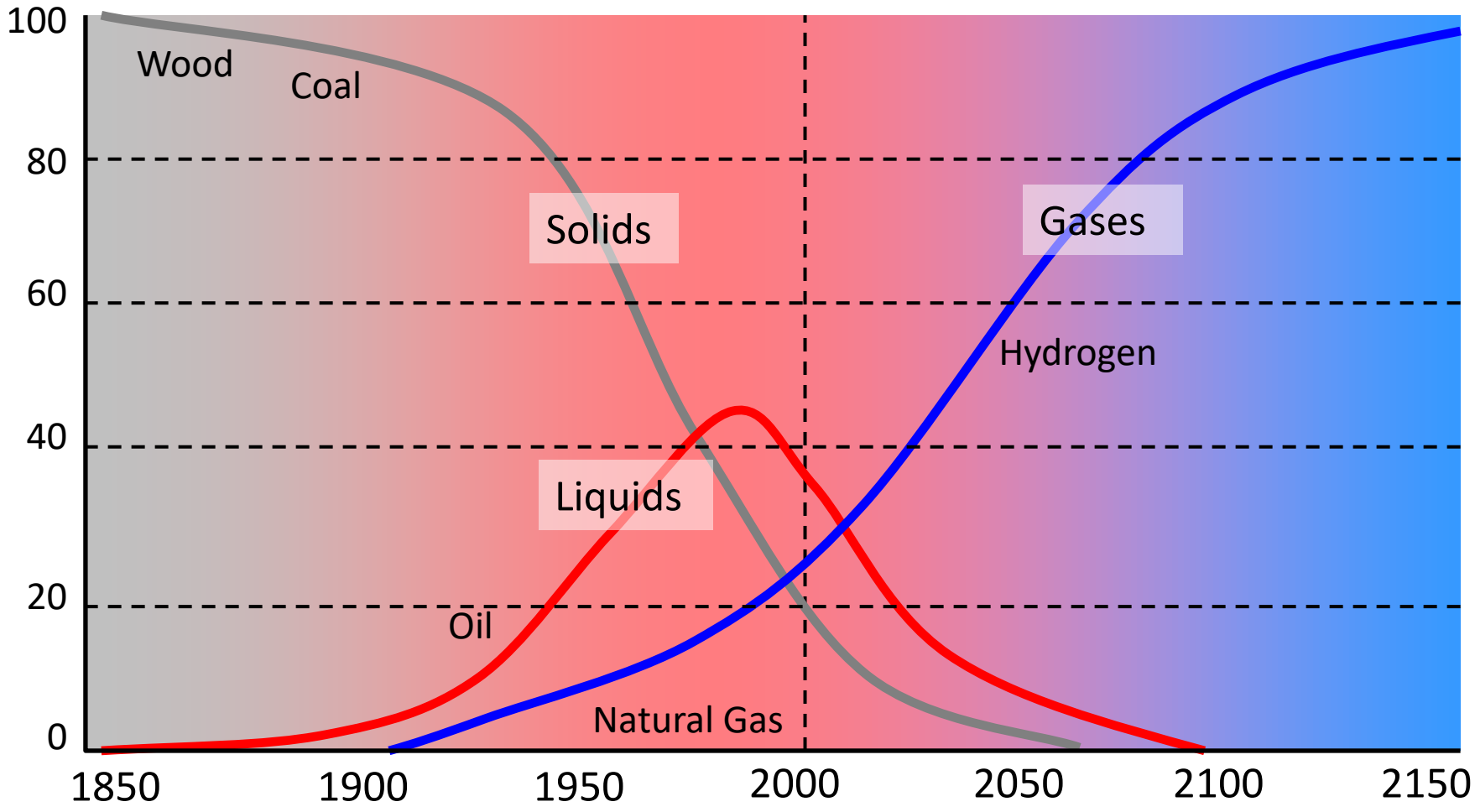
- From the above scenario it is clear that our life style and standard of living are closely related to energy needs.

Percapita Energy Use

- The figure shows the correlation between percapita energy use and GNP (Gross National Product).
- The developed countries like U.S.A, Japan, Switzerland, etc., with high GNP show high energy use while the poor countries like India, China, Ethiopia with low GNP show low energy use



Global Energy Systems Transition, (% of market)



RENEWABLE AND NON-RENEWABLE ENERGY RESOURCES

Based on continual utility, natural resources can be classified into two types.

1. Renewable energy resources.
2. Non-renewable energy resources.

Renewable energy resources (or) Non-Conventional energy resources

Renewable energy resources are natural resources which can be regenerated continuously and are inexhaustible. They can be used again and again in an endless manner.

Merits of renewable energy resources

1. Unlimited supply.
2. Provides energy security.
3. Fits into sustainable development concept.
4. Reliable and the devices are modular in size.
5. Decentralized energy production.

2 Non-renewable energy resources (or) conventional energy sources

Non-renewable energy resources are natural resources, which cannot be regenerated once they are exhausted. They cannot be used again.

Examples: Coal, petroleum, natural gas and nuclear fuels.

Wood is renewable resources but not coal-why?

Wood is renewable resources because we can get new wood by growing a sapling into a tree within 15-20 years.

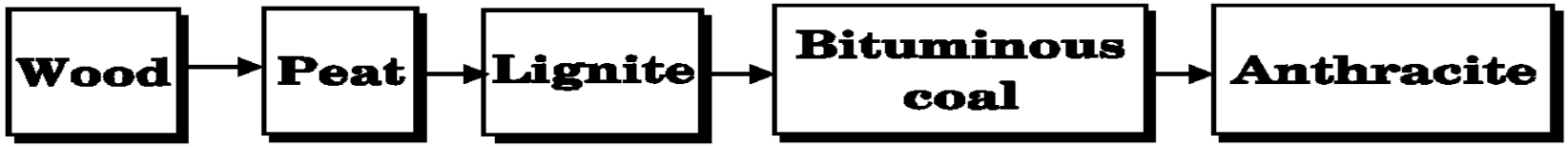
But the formation of coal from trees has taken million of years and cannot be regenerated in our life time.

NON-RENEWABLE ENERGY SOURCES

Coal

Coal is a solid fossil fuel formed in several stages as buried remains of land plants that lived 300-400 million year ago, were subjected to intense heat and pressure over millions of years.

The various stages of coal during the coalification of wood is



The carbon content of Anthracite is 90% and its calorific value is 8700 k.cal. The carbon content of bituminous, lignite and peat are 80, 70 and 60% respectively.

India has about 5% of worlds coal. Indian coal is not good because of poor heat capacity.

Disadvantages of using coal

1. When coal is burnt it produces CO_2, causes global warming.
2. Since coal contains impurities like S and N, it produces toxic gases during burning.

Petroleum

It is the lifeline of global economy. There are 13 countries in the world having 67% of the petroleum reserves which together form the OPEC (Organization of Petroleum exporting countries). About 1/4th of the oil reserves are in Saudi Arabia.

Petroleum or crude oil is a thick liquid consisting of more than hundreds of combustible hydrocarbons with small amount of S, O, N as impurities.

Occurrence

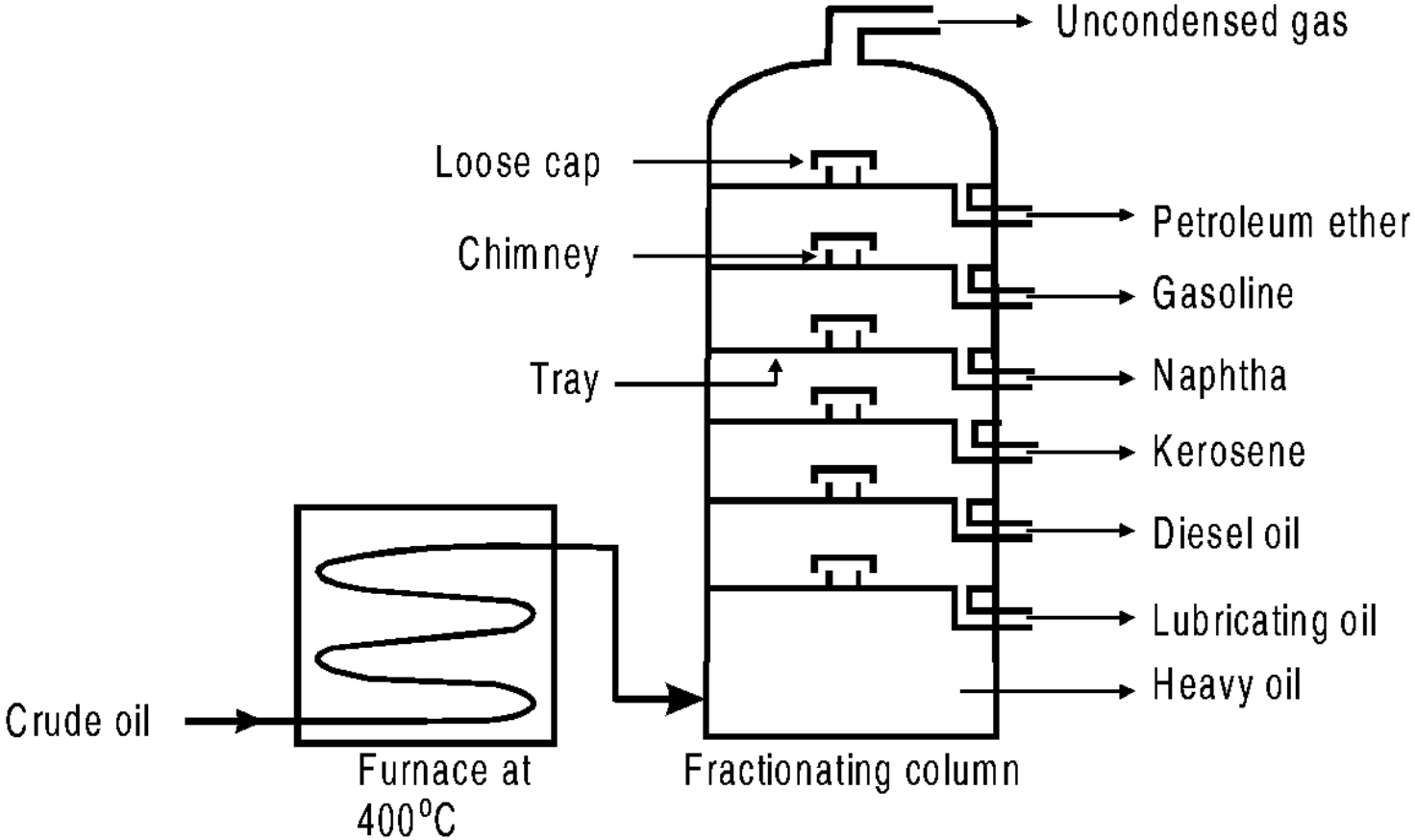
The fossil fuel (coal and petroleum) was formed by the decomposition of dead animals and plants, that were buried under lake and ocean at high temperature and pressure for million of years.

Fractional distillation

From the crude-petroleum oil, the various hydrocarbons are separated by purifying and fractionating the crude petroleum oil.

We get a large variety of products from this, namely, petroleum gas, kerosene, petrol, diesel, fuel oil, lubricating oil, paraffin wax, asphalt, plastic etc.

Fractionating column



Petroleum is a cleaner fuel as compared to coal as it burns completely and leaves no residue. It is also easier to transport and use. That is the reason why petroleum is preferred amongst all the fossil fuels

LPG (Liquefied petroleum gas)

Petroleum gas, obtained during cracking and fractional distillation, can be easily converted into liquid under high pressure as LPG. LPG is colourless and odourless gas.

But during bottling some mercaptans is added, which produces bad odour, thereby any leakage of LPG from the cylinder can be detected instantaneously.

Natural gas is found above the oil in oil well. It is a mixture of 50-90% methane and small amount of other hydrocarbons.

Its calorific value ranges from 12,000-14,000 k . cal/m^3

Natural gases are of two types.

1. Dry gas: If the natural gas contains lower hydrocarbons like methane and ethane, it is called dry gas.

2. Wet gas: If the natural gas contains higher hydrocarbons like propane, butane along with methane it is called wet gas.

Like petroleum oil, natural gas is also formed by the decomposition of dead animals and plants, that were buried under lake and ocean, at high temperature and pressure for million of years.

Energy management and conservation

Energy management

- Energy management includes planning and operation of energy production and energy consumption units as well as energy distribution and storage.

- For the past decades energy generation has been shifted to alternative energy sources like renewable energy forms such a solar wind and biomass energy.

- Apart from the growth in the energy sector there has been an equivalent increase in business and Organisations.

- Energy Management introduced in an organisation can effectively manage how much energy they produce and how to control and monitor.

- Energy and money can be saved by employing Energy Management principles.

Aims of Energy Management:

- ✓ Resource conservation
- ✓ Climate protection
- ✓ Cost savings minimise waste
- ✓ Minimise Environmental effects

- ✓ The Ultimate aim or objectives of this process is not only to save the cost but also to achieve complete environmental sustainability

Principles of Energy Management:

- Reducing needless energy use
- Increasing energy efficiency
- Purchasing energy at a lesser cost
- Changing operations to allow for lower cost energy purchases

Following five steps are important in the process of Energy Management

- Step 1 -Collecting and analysing continuous data
- Step 2- Identify optimisation in equipment schedules
- Step 3 -Calculate return on investment
- Step 4 -Execute energy optimization solutions
- Step 5 -Repeat step 2 to continue optimising Energy Efficiency

Importance of Energy management:

- ✓ Primary goal of energy management is design and maintain optimal energy consumption
- ✓ Energy Management is a best strategy for lowering direct and immediate energy use
- ✓ Managing and reducing energy usage not only results in financial savings but also benefits the environment and a company's reputation.
- ✓ Energy should be considered a business cost.

ENERGY CONSERVATION

- Energy conservation is the practice of using less energy in order to lower the cost and reduce environmental impact.
- This can be achieved by
  - energy more efficiency (using less energy for a constant service)
  - Reducing the amount of service

Needs to save energy:

✓ We have finite and our demand for energy is growing.

✓ Most fossil feels likely to run out and forcing which to alternative sources of energy.

Aims of energy conservation

- To reduce overall energy demand
- To lower energy cost
- To reduce energy consumption
- To lower the overall Greenhouse gas emission

Principle or law of conservation of energy

- The principle of energy conservation states that the energy can neither be created nor destroyed but it can be transformed from one type to another
- According to this the total energy of an isolated system remains constant

Importance of energy conservation

Energy conservation is very important because of the following reasons.

- ✓ It reduces our usage of non-renewable energy resources
- ✓ It also helps to save money on energy costs including utility bills and other energy bills
- ✓ It also cuts doen on expanding devolopment, where natural resource extraction is impacting natural areas
- ✓ It insists us to replace the energy, used with an alternative energy sources
- ✓ When we conserve the energy more efficiently, we direcltly reduce the amount of green house gas emissions entering into the earths atmosphere.

15 ways to conserve energy:

1. **Adjust your day today behaviours:** switch off lights and appliances when you do not need them

2. **Replace your light bulbs:** traditional incandescent light bulbs consume more amount of electricity and must be replaced by energy efficient like LED bulbs.

3. **Use smart power strips :** the electricity used by electronics, when they are turned off or stand by mode, are the major source energy waste. Smart power strips eliminates this problem, when they are not in use.

4. **Install Programmable or smart thermostat:** It automatically turn –off or reduce heating and cooling during the time when you are asleep or away

15 ways to conserve energy:

5. **Purchase energy efficient appliances:** this will consume less energy during use.

6. **Reduce your water heating expenses:** this will reduce the water heater expenses from 8 – 300% as compared to conventional water heaters

7. **Install energy efficient Windows:** prevent heat loss through windows

8. **Update your HVAC systems** at your homes.

9. **Weatherize your home** – air leaks in to yopur home are windows, doors and vents

15 ways to conserve energy:

10. **Insulate your home:** Insulation retains heat during winter

11. Wash your cloths in cold water

12. Replacing dirty air filters in HVAC and AC's

13. Microwave is more energy efficient, microwave oven can be used instead of ordinary stove.

14. Using Natural light like sun we can reduce the energy consumption.

15. Dress appropriately for the weather inside and outside.

New energy sources, Need of new sources

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Need for new energy sources

✓Fossil fuels and nuclear energy are the important resources used to meet most of our energy needs today these are expected to be widely used in the near future.

✓ Fossil and a nuclear energy resources are non-renewable and will someday be exhausted, while their continued use possess environmental risks related to air pollution, land use and waste disposal.

✓ These issues have stimulated the research for new energy sources for producing and using energy.

Geo suitability of establishing renewable energy sources

The geographic suitability of renewable energy resources varies depending on factors such as climate, topography, and local environmental conditions.

A brief overview of the suitability of different renewable energy resources in various geographic regions is given below

1. **Solar Energy:**

- ✓ Solar energy is most abundant in regions with high levels of sunlight, such as deserts and areas near the equator.

- ✓ Countries with vast expanses of desert, like the southwestern United States, North Africa, and the Middle East, are particularly well-suited for large-scale solar energy projects.

- ✓ However, solar energy can still be harvested effectively in regions with less sunlight through the use of photovoltaic panels, though the efficiency may vary.

Geo suitability of establishing renewable energy sources

2. Wind Energy:

- Wind energy is most viable in regions with consistent, strong winds, such as coastal areas, plains, and mountain passes.
- Coastal regions and offshore areas tend to have particularly strong and consistent wind resources, making them ideal locations for offshore wind farms.
- Countries with large coastlines or extensive flat terrain, such as the United States, China, and parts of Europe, have significant wind energy potential.

3. Hydroelectric Power:

- Hydroelectric power relies on the availability of flowing water, making regions with rivers, streams, and waterfalls well-suited for this renewable energy source.
- Mountainous regions with ample rainfall and snowmelt, such as the Alps, Andes, and Himalayas, often have abundant hydroelectric potential.
- However, the environmental impact of large-scale hydroelectric projects, such as dams, must be carefully considered.

Geo suitability of establishing renewable energy sources

4. Biomass Energy:

- Biomass energy can be generated from organic materials such as agricultural residues, forest residues, and dedicated energy crops.
- Regions with significant agricultural or forestry industries can utilize these organic materials to produce biomass energy.
- Additionally, urban areas with ample organic waste can also benefit from biomass energy production through technologies like anaerobic digestion and incineration.

5. Geothermal Energy:

- Geothermal energy relies on heat from the Earth's interior, making it most accessible in regions with high geothermal activity, such as tectonic plate boundaries and volcanic areas.
- Countries located along the Pacific Ring of Fire, such as Iceland, the United States (particularly in the western states), and parts of East Africa, have significant geothermal energy potential.
- However, even in regions with lower levels of geothermal activity, technologies like ground-source heat pumps can still harness geothermal energy for heating and cooling purposes.

Geo suitability of establishing renewable energy sources

Overall, the geographic suitability of renewable energy resources depends on a combination of factors, and each region may have its own unique mix of viable options.

Advances in technology and infrastructure can also expand the potential for renewable energy development in previously less suitable areas.

Different types of new energy sources (or) renewable energy resources

1. Hydrogen energy
2. Ocean Thermal Energy conversion
3. Tidal and wave energy
4. Geothermal energy
5. Solar energy
6. Wind energy
7. Biomass energy

Hydrogen energy and its applications

Hydrogen energy:

- ✓ Hydrogen energy is a type of renewable energy that uses hydrogen gas as a fuel to generate electricity or power vehicles.

- ✓ This fuel that has potential of being widely used in the future is hydrogen gas. Like a natural gas, H2 can be burned to heat buildings, cook food and produce electricity in power plants

- ✓ It has high calorific value

- ✓ It non-polluting energy source

$$2H_2 + O_2 \longrightarrow 2H_2O + 150\ kJ$$

Sources of hydrogen

- ✓ Hydrogen is the most abundant element in the universe, and it can be obtained from a variety of sources such as natural gas, biomass, and water.
- ✓ The process of generating hydrogen involves separating it from other elements in a process called electrolysis.
- ✓ This process uses electricity to split water molecules into hydrogen and oxygen.
- ✓ Microbes such as green algae, consume water in the presence of sun light and produce H_2 as a byproduct.
- ✓ Ethanol reacts with high –temperature stream to produce H_2
- ✓ Once the hydrogen is separated, it can be stored and transported to be used as a fuel.
- ✓ One of the main advantages of hydrogen energy is that it produces zero emissions, as the only by product of hydrogen combustion is water vapour.

Applications of Hydrogen Energy:

- ✓ Hydrogen is a reagent used in many industries including chemicals, textile fibre manufacturing, glass electronic and metallurgy.
- ✓ It is also used as a fuel for rocket launchers.
- ✓ In electronics hydrogen is used as a carrier gas, for manufacture of electronic components.
- ✓ In industry hydrogen combines with nitrogen to produce ammonia, base for fertilizers.
- ✓ It is a good reagent for textile fibres like nylon.

Applications of Hydrogen Energy:

- ✓ Hydrogen is used in metallurgy for heat treatment process mechanical parts or to alter their properties.
- ✓ Atomic hydrogen welding (AHW) is a type of arc welding which utilizes hydrogen environment.
- ✓ A mixture of hydrogen and nitrogen is used to prevent oxidation in flat glass production.
- ✓ Hydrogen fuel cell uses hydrogen as a fuel in an electrochemical process to produce electrical energy with water and heat.

Two main applications of hydrogen fuel cells are

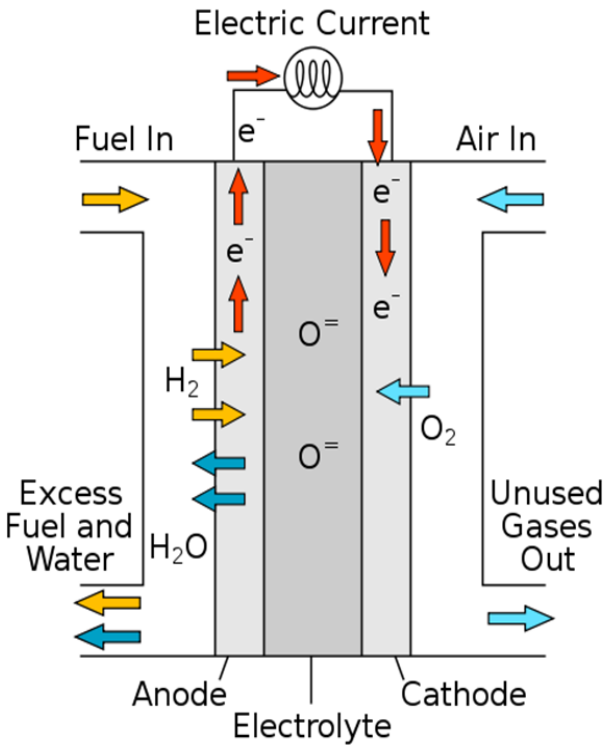
- **Stationery power sources** are used to power office buildings, data centers, grocery stores and telecommunication towers, it is used as a part of UPS system.

- **Hydrogen fuel cell vehicles (FCVs)** are a type of electric vehicle that use hydrogen gas as their primary fuel source.

- The hydrogen is stored in high-pressure tanks and is combined with oxygen from the air in a fuel cell to generate electricity.

- The electricity is then used to power an electric motor, which drives the vehicle's wheels.

- Hydrogen fuel cell trains have now appeared. Hydrogen fuel cells are found a number of marine applications.



Advantages of hydrogen energy:

- Hydrogen is readily available
- It does not produce harmful emissions
- It is environmentally friendly
- Efficient and more powerful than fossil fuels
- It is renewable
- It reduces carbon footprints
- It can be used for long time

Disadvantages of hydrogen energy:

- The production of hydrogen is highly energy-intensive, requiring large amounts of energy and **expensive technologies**

- **Storage and Transportation:** Unlike traditional fuels, hydrogen is lightweight and highly volatile. It requires specialized storage and transportation Highly inflammable

- **Safety concerns:** Hydrogen is highly flammable, which raises serious safety concerns. Any leakage or exposure to oxygen can potentially result in an explosion.

- **Limited availability:** Hydrogen is not readily available at gas stations and service stations, making it difficult to refuel vehicles.

Concept, origin, power plants and applications of Ocean energy resources

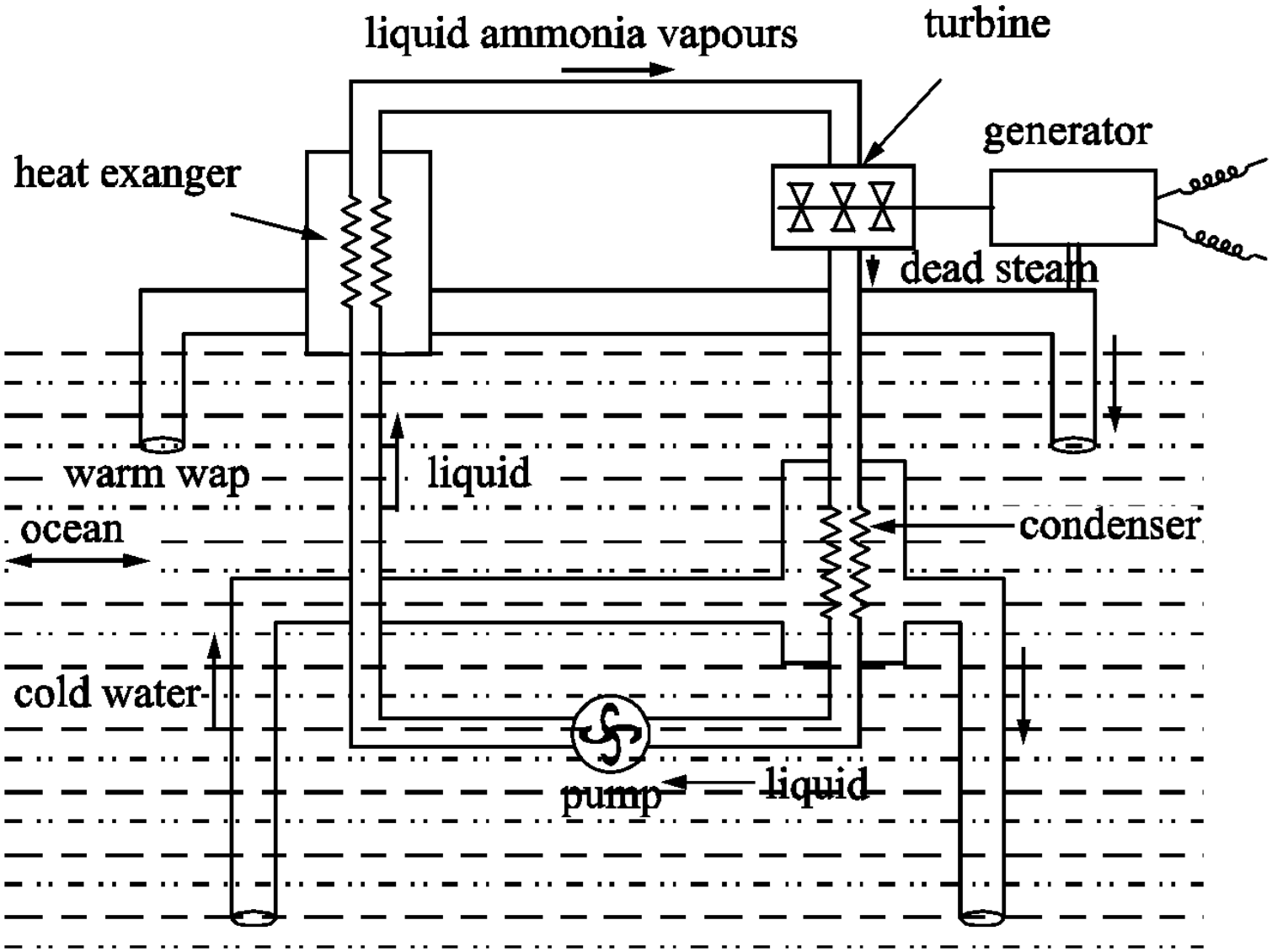
Ocean Thermal Energy (OTE) conversion

There is often large temperature difference between the surface level and deeper level of the tropical oceans. This temperature difference can be utilized to generate electricity. ***The energy available due to the difference in temperature of water is called ocean thermal energy***.

Condition

The temperature difference should be of 20°C or more is required between surface water and deeper water.

Ocean thermal energy



The warm surface water of ocean is used to boil a low boiling liquid like ammonia. The high vapour pressure of the liquid, formed by boiling, is then used to turn the turbine of the generator and generates electricity. The cold water from the deeper ocean is pumped to cool and condense the vapour into liquid.

Significance of OTE

1. OTE is continuous, renewable and pollution free.
2. The use of cold deep water, as the chiller fluid in air-conditioning, has also been proposed.
3. Electric power generated by OTE can be used to produce hydrogen.

Tidal energy conversion

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- Tidal energy is a form of renewable energy that harnesses the power of the tides to generate electricity.

- Tides are caused by the gravitational pull of the moon and the sun, and as the water moves in and out with the tides.

- Tidal energy conversion is the process of generating electricity from the movement of ocean tides.

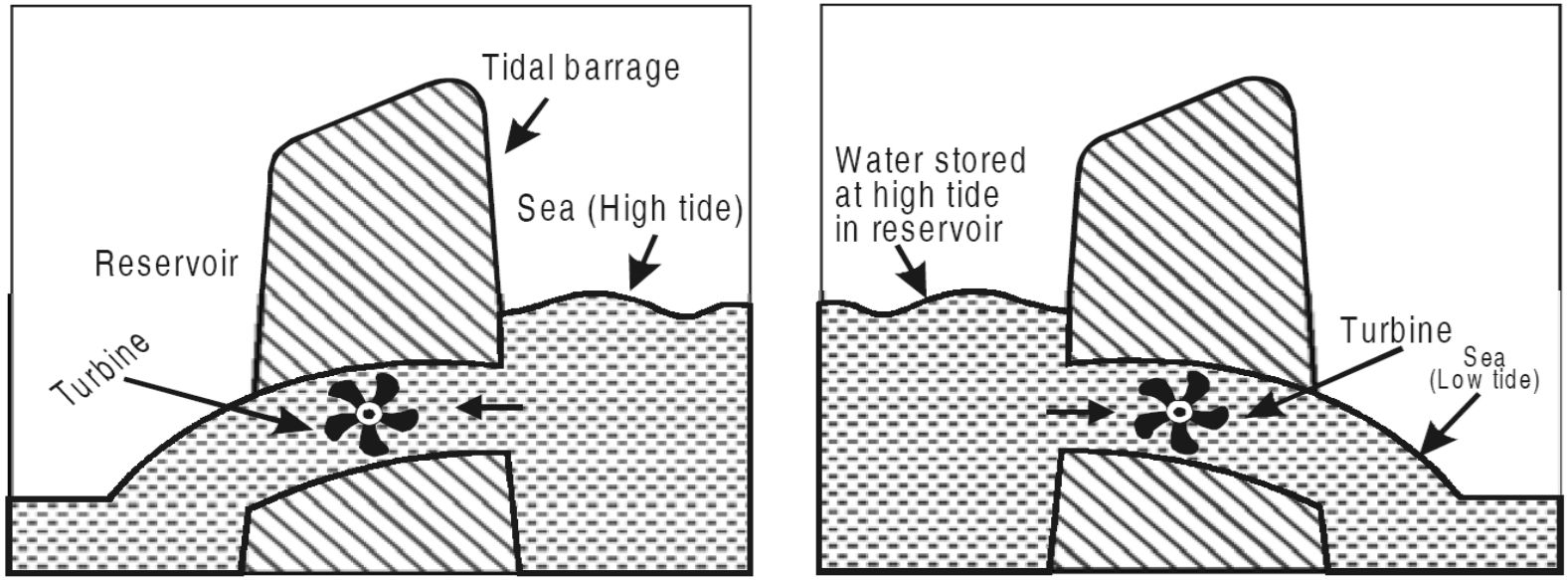
- This is typically done by placing turbines in tidal currents or using underwater turbines that are driven by the kinetic energy of moving water.

- As the tides move in and out, the turbines spin and generate electricity, which can then be used to power homes, businesses, and other electrical devices.

- There are two main types of tidal energy systems: tidal stream and tidal barrage.

- ✓ **Tidal stream** systems use underwater turbines that are placed in areas with strong tidal currents, such as narrow channels or between islands. The turbines are turned by the movement of the water, generating electricity that can be transmitted to shore.

- ✓ **Tidal barrage systems**, on the other hand, use large dams or barrages to trap the incoming tide and then release it through turbines as the tide goes out. This method is similar to hydroelectric power generation, but uses the power of the tides rather than the flow of a river



Water flows into the reservoir from sea. **Water flows out from the reservoir to the sea.**

Significance of tidal energy

1. Tidal power plants do not require large valuable lands as they are on the bay or areas of estuaries
2. As the sea water is inexhaustible, it is completely, it is independent of the uncertainty of precipitation (rainfall)
3. It is pollution-free energy source as it does not use any fuel and also does not produce any waste

APPLICATIONS OF TIDAL ENERGY

- Electricity can be generated from the tidal energy
- Tidal energy is used to grinning mills for the mechanical crushing of grains.
- Tidal energy is used to rotate at turbine
- Tidal energy is used to store energy in hydroelectric Dam acting as large energy storage
- Tidal barrage and reservoirs can be modified to store energy
- Tidal barrages are capable of preventing damages to the coast during high storms
- Tidal barrage is also help to create easy transport.

Advantages of tidal energy

- ✓ It is environment friendly
- ✓ It is cheaper, efficient and predictable
- ✓ Low operating and maintenance cost
- ✓ Protects coastal flooding
- ✓ Power output is highly predictable

Disadvantages of tidal energy

- ➢ Construction of title Power Plant is expensive.
- ➢ Equipment maintenance difficult
- ➢ Storage capacity is required.
- ➢ Negative influence on marine life forms.
- ➢ Location Limited

Concept, origin and power plants of geothermal energy.

Geothermal Energy (GTE) and types of geothermal power plants

Geothermal energy is the energy that is derived from the heat within the earth's crust. The earth's crust contains a large amount of heat, which is generated by the radioactive decay of minerals and the residual heat left over from the earth's formation. This heat can be used to generate electricity through the use of geothermal power plants.

- GTE is used for space heating and cooling
- GTE is used to generate electricity
- It is also used for industrial process heat
- geothermal heat pumps are used to heat buildings in the winter and cool them in summer

CONCEPT

✓Geothermal technology extracts the heat found within the subsurface of the earth, which can be used directly for heating and cooling or converting it to electricity.

✓The steam comes from the reservoirs of hot water found a few miles below the earth surface rotate a turbine that activates a generator, which produces electricity.

POWER PLANTS OF GEOTHERMAL ENERGY:

✓Geothermal power plant uses hydro thermal resources that have both water and heat.

✓Geothermal power plants require high temperature hydro thermal resources that come from either dry steam wells or from hot water wells.

✓Generally, we use this resources by drilling wells into the earth and then piping steam or hot water to the surface.

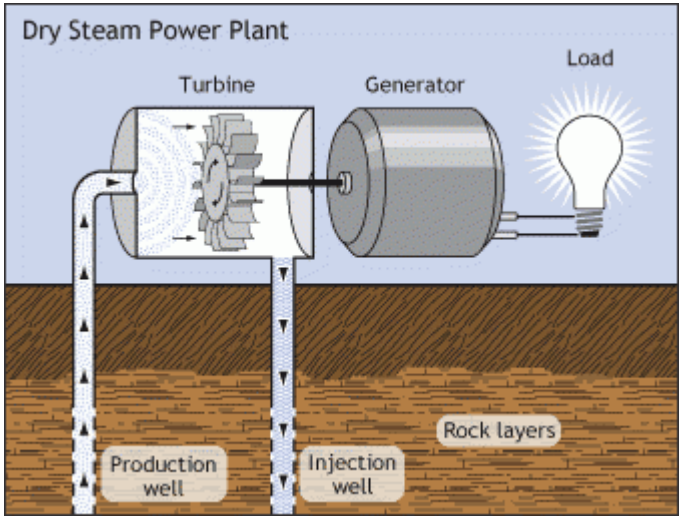
✓The hot water or steam rotates a turbine that generates electricity, the depth of the geothermal Wells is as much as two miles.

TYPES OF GEOTHERMAL POWER PLANTS:

There are 3 basic types of geothermal power plants

1) Dry steam power plant

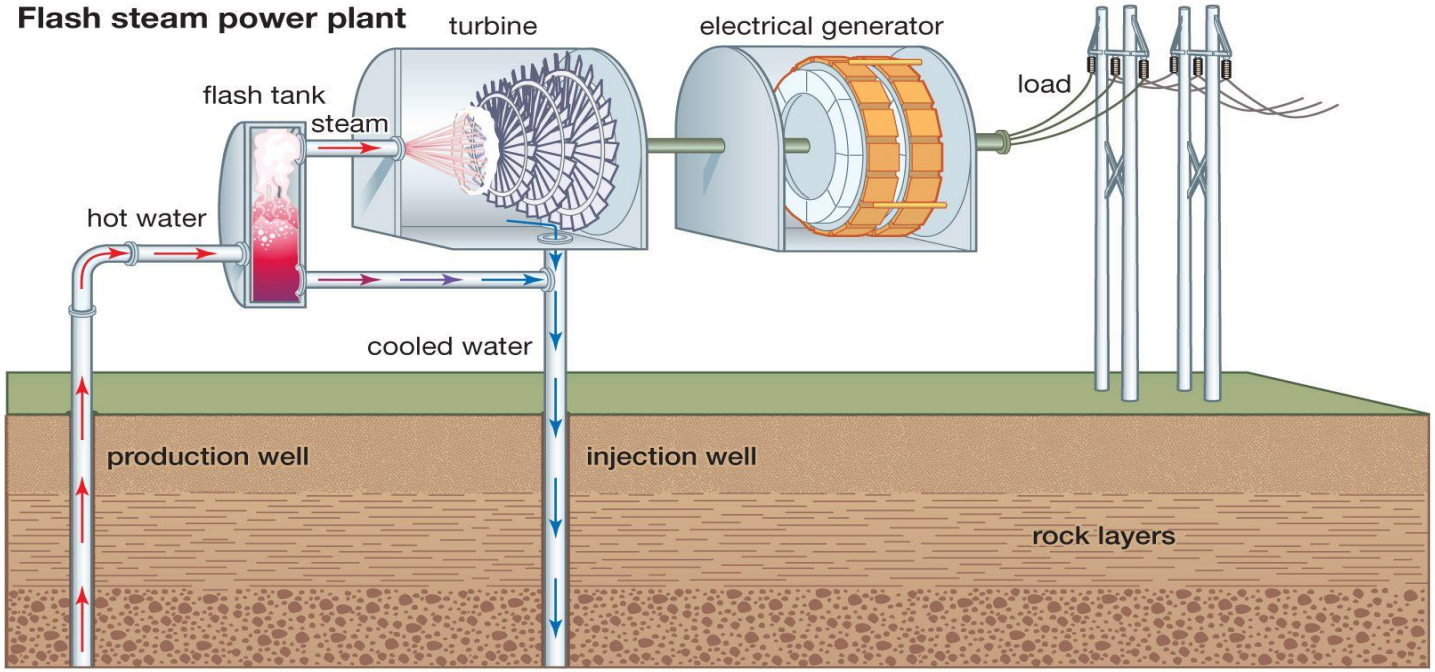
- ✓ It uses steam directly from here geothermal reservoir to drive generator's turbine
- ✓ The steam goes directly to a turbine, which drives a generator that produces electricity.
- ✓ The steam eliminates the need to burn fossil fuels to run the turbine. (Also eliminating the need to transport and store fuels!)
- ✓ This is the oldest type of geothermal power plant. It was first used at Lardarello in Italy in 1904.
- ✓ These plants emit only excess steam and very minor amounts of gases.



2) Flash steam power plant

- It takes high pressure hot water from deep inside the earth and Converts it into steam to drive generator's turbine.

- When the steam cools, it condenses to water and is injected back into the ground to be used again. Most thermal power plants are flash steam plants.



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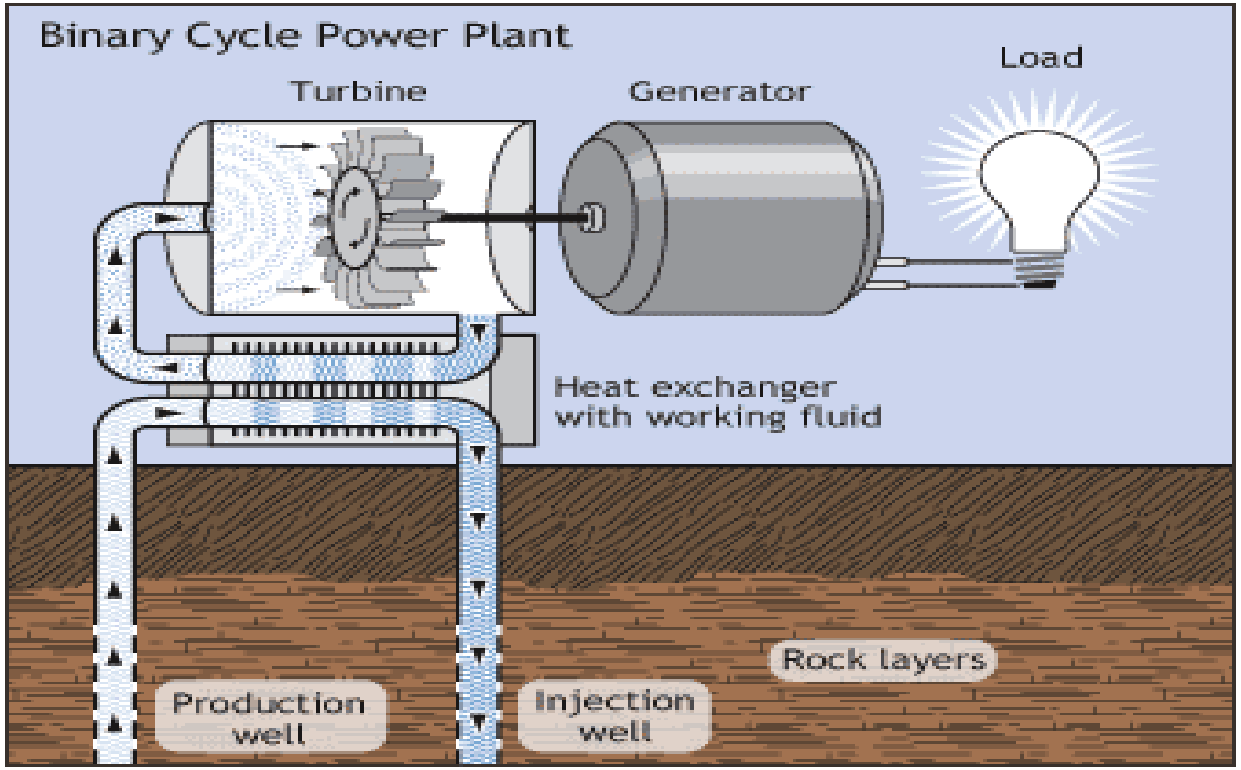
3) Binary cycle power plants

- It transfers the heat from geothermal hot water to another liquid.

- The heat causes the second liquid to convert it into steam which is used to drive a generator's turbine.



Advantages of geothermal energy:

- GT energy is environmentally friendly
- GT is a source of renewable energy
- The potential of GTE is huge
- GTE is natural

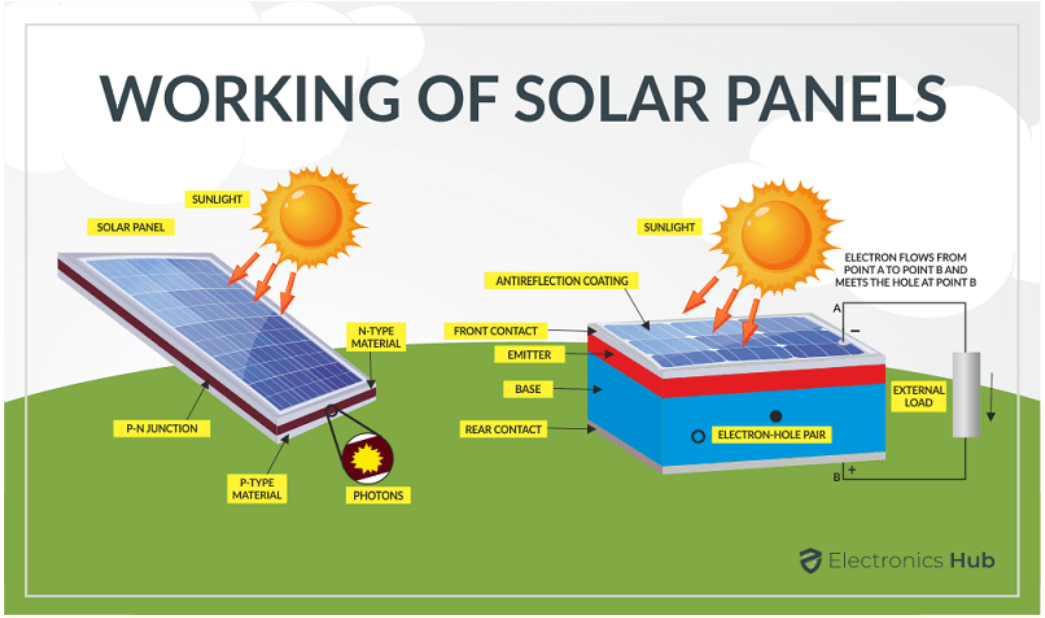
Disadvantages

- Location is restricted
- There are many other gases released into the atmosphere
- May cause earthquakes
- It is expensive resource, management is required to maintain sustainability

Solar energy

The energy that we get directly from the sun is called solar energy.

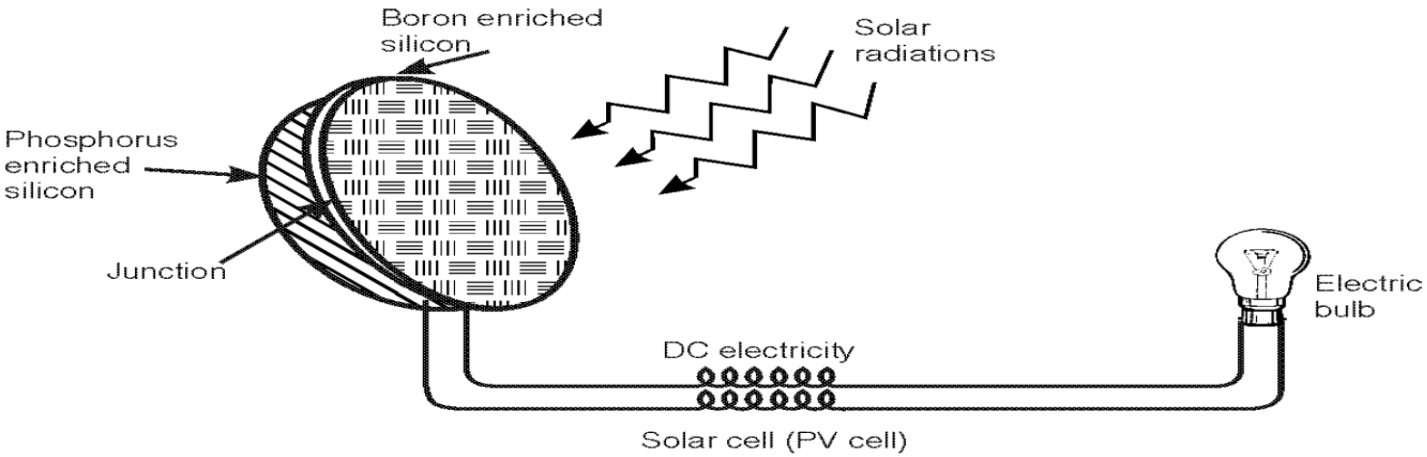
The nuclear fusion reactions occurring inside the sun release enormous amount of energy in the form of heat and light. Several techniques are available for collecting, converting and using solar energy.



Methods of Harvesting Solar Energy

Some important solar energy harvesting devices are given below.

1. Solar cells (or) photovoltaic cells (or) PV cells



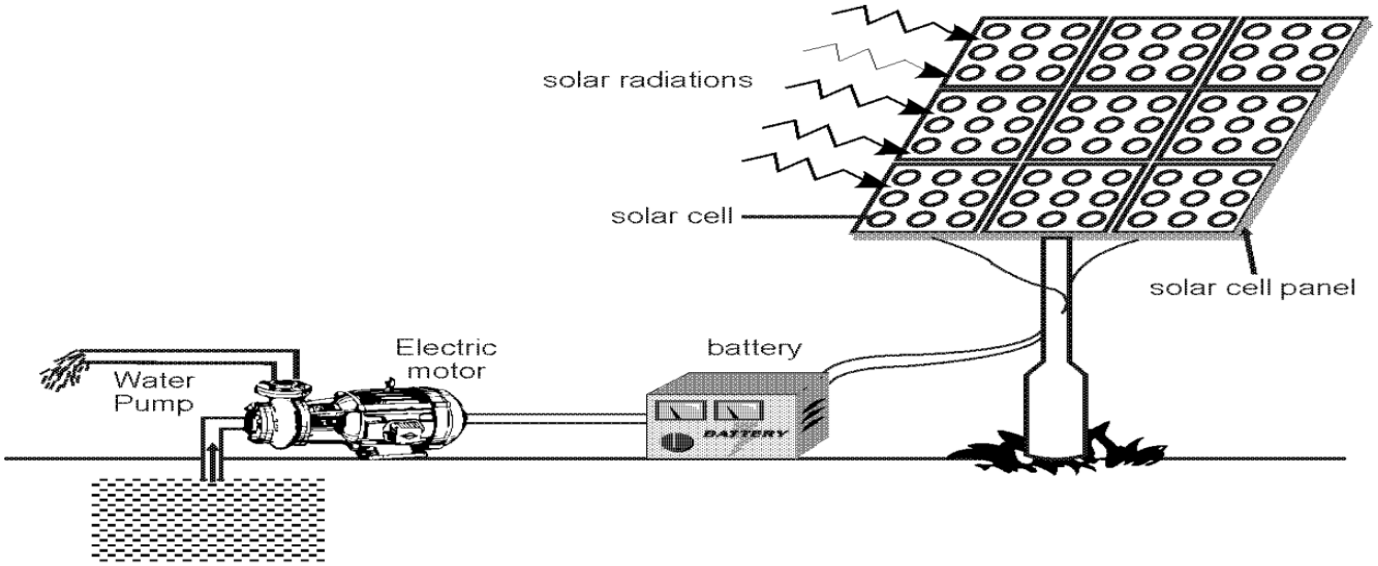
Solar cells consists of a p-type semiconductor (such as Si doped with B) and n-type semiconductor (such as Si doped with P). They are in close contact with each other. When the solar rays fall on the top layer of p-type semiconductor, the electrons from the valence band get promoted to the conduction band and cross the p-n junction into n-type semiconductor. There by potential difference between two layers is created, which causes flow of electrons (ie., an electric current).

Uses

Used in calculators, electronic watches, street lights, water pumps to run radios and TVs.

Solar Battery

When a large number of solar cells are connected in series it form a solar battery. Solar battery produce more electricity which is enough to run water pump, to run street-light, etc., They are used in remote areas where conventional electricity supply is a problem.



2. Solar Heat Collectors

Solar heat collectors consist of natural materials like stones, bricks (or) materials like glass, which can absorb heat during the day time and release it slowly at night.

Uses - It is generally used in cold places, where houses are kept in hot condition using solar heat collectors.

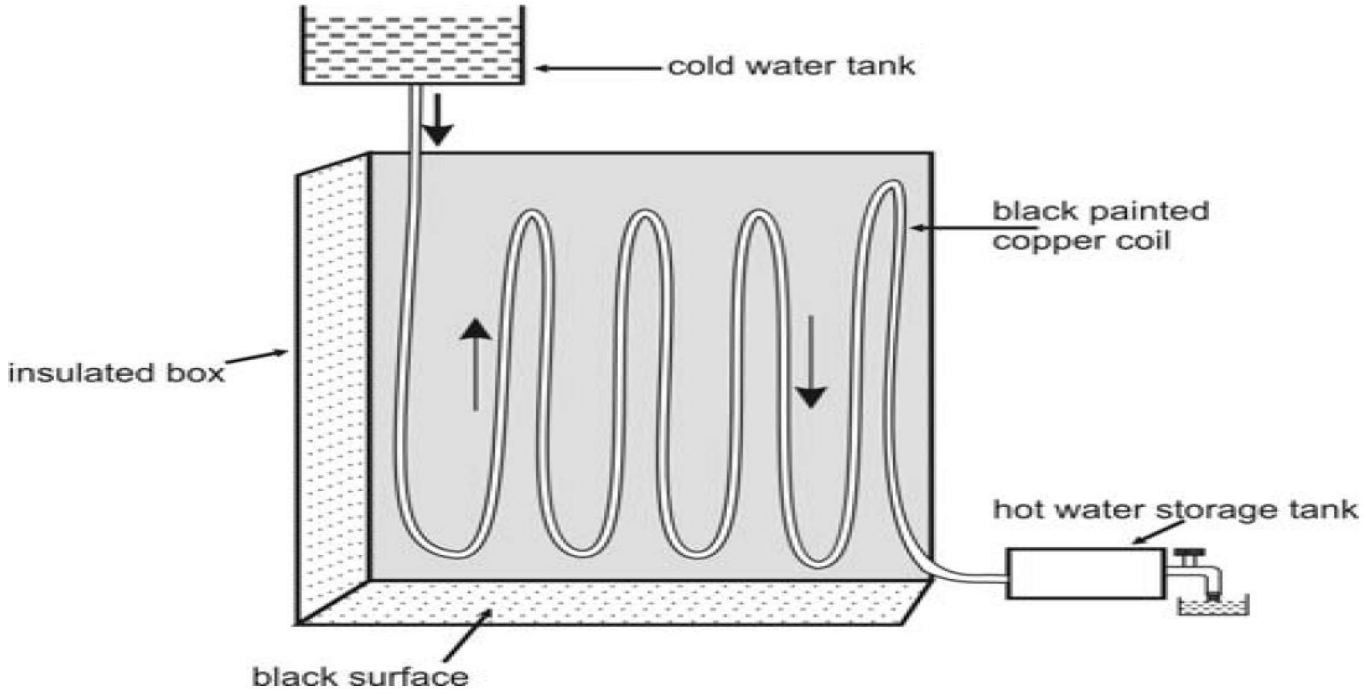
3. Solar water heater

It consists of an insulated box inside of which is painted with black paint.

It is also provided with a glass lid to receive and store solar heat.

Inside the box it has black painted copper coil, through which cold water is allowed to flow in, wl1ich gets heated up and flows out into a storage tank.

From the storage tank water is then supplied through pipes.



Significance of solar energy

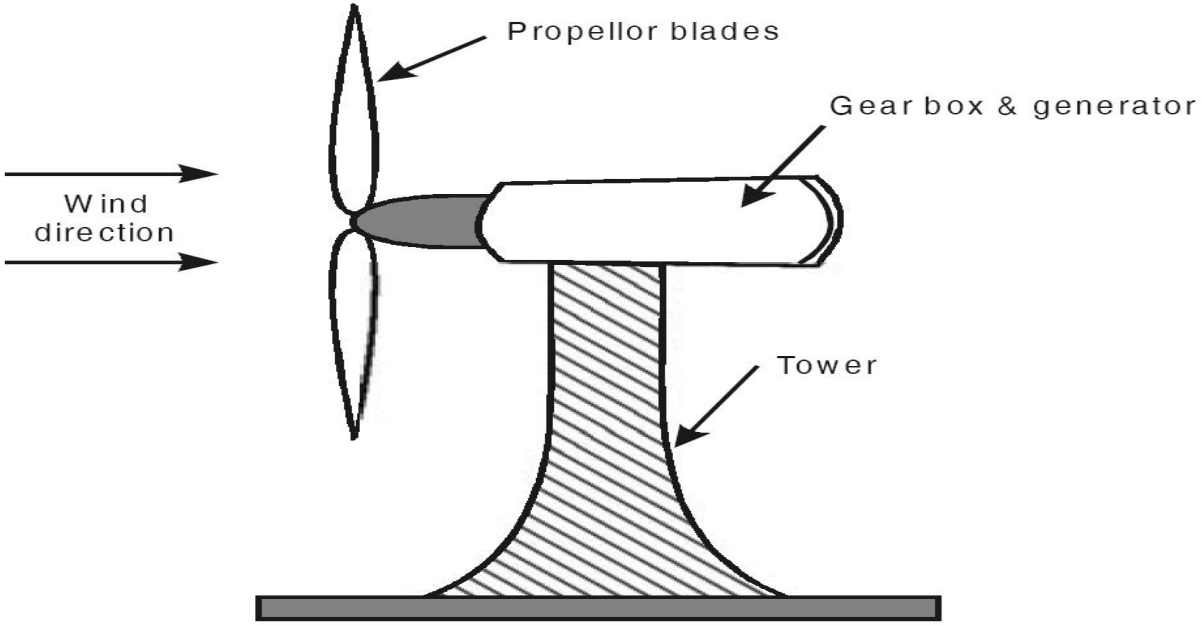
1. Solar cells are noise and pollution free.
2. Solar water heaters, cookers, require neither fuel nor attention while cooking food.
3. Solar cells. can be used in remote and isolated areas, forests, hilly regions.

Wind energy

Moving air is called wind. Energy recovered from the force of the wind is called wind energy. The energy possessed by wind is because of its high speed. The wind energy is harnessed by making use of wind mills.

1. Wind Mills

The strike of blowing wind on the blades of the wind mill make it rotating continuously. The rotational motion of the blade drives a number of machines like water pump, flour mills and electric generators.



2. Wind Farms.

When a large number of wind mills are installed and joined together in a definite pattern ir forms a wind farm. The wind farms, produce a large amount of electricity

The minimum speed required for satisfactory working of wind generator is 15 km/hr.

Advantages

1. It does not cause any pollution.
2. It is very cheap

Significance of wind energy

1. The generation period of wind energy is low and power generation starts from commissioning.
2. It is recommended to broaden the nation's energy options for new energy sources.
3. It is made available easily in many off-shore, on-shore and remote areas.

Biomass energy

Biomass is the organic matter, produced by plants or animals, used as sources of energy. Most of the biomass is burned directly for heating, cooling and industrial purposes.

Wood, crop residues, seeds, cattle dung, sewage, agricultural wastes, etc.,

1. Biofuels

Biofuels are the fuels, obtained by the fermentation of biomass.

Ethanol, methanol.

(a) Ethanol: Ethanol can be produced from the sugarcane. Its calorific value is less when compared to petrol, and produces much less heat than petrol.

(b) Methanol: Methanol can be easily obtained from ethanol or sugar-containing plants. Its calorific value is also too low when compared to gasoline and diesel.

(c) Gasohol: Gasohol is a mixture of ethanol + gasoline. In India trial is being carried out to use Gasohol in cars and buses.

Uses of Alternative source of energy

Need of Alternate (Renewable) Energy Sources (or) Role of Alternate (Renewable) Energy sources in environmental impact

1. The importance of solar energy can be emphasized particularly in view of the fact that fossil fuels and other conventional sources are not free from environmental implications.
2. Energy sources which have least pollution, safety and security snags and are universally available have the best enhance of large scale utilization in future.
3. Hydro-electric power generation is expected to upset the ecological balance existing on earth.
4. Besides space heating, hydel power plants critically pollute the aquatic and terrestrial biota.

5. Radioactive pollutants released from nuclear power plants are chronically hazardous. The commissioning of boiling water power reactors (BWRS) have resulted in the critical accumulation of large number of long lived radionuclides in water.
6. The dangerous radiowaste cannot be buried in land without the risk of polluting soil and underground water. Nor the waste can be dumped into the rivers without poisoning aquatic life and human beings as well.
7. The burning of coal, oil, wood, dung cakes and petroleum products have well debated environmental problems. The smoke so produced causes respiratory and digestive problems leading to lungs, stomach and eye diseases.
8. The disposal of fly ash requires large ash ponds and may pose a severe problem considering the limited availability of land. So, the non conventional sources of energy needed.

Examples for diffrent energy conversion

Energy Sources	Types of energy	Energy after conversion
Hydro (water)	Potential Energy	Kinetic Energy → Electrical energy.
Wind, Hydro	Kinetic Energy	Electrical Energy.
Geothermal, Ocean-thermal	Thermal Energy	Mechanical Energy → Electrical Energy.
Solar	Radiant Energy	Thermal Energy → Electrical Energy.
Oil, Coal-gas, Biomass	Chemical Energy	Thermal Energy → Electrical Energy.
Uranium, Thorium	Nuclear Energy	Thermal Energy → Mechanical Energy → Electrical Eneregy.

Measures recommended for conservation of energy resource

1. Conservation of Energy

1. Switch off lights, fans and other appliances when not in use.
2. Use solar heater for cooking your food on sunny . days, which will cut down your LPG expenses.
3. Dry the clothes in sunlight instead of driers.
4. Grow trees near the houses and get a cool breeze and shade. This will cut off your electricity charges on AC and coolers.
5. Use always pressure cooker.
6. Ride bicycle or just walk instead of using car and scooter

ROLE OF AN INDIVIDUAL IN CONSERVATION OF ENERGY RESOURCES

Individuals play a significant role in the conservation of energy resources through their daily choices and habits. Here are some ways individuals can contribute to energy conservation:

1. **Energy-Efficient Appliances**: Use energy-efficient appliances and lighting fixtures to reduce electricity consumption.
2. **Smart Energy Use**: Turn off lights, electronics, and appliances when not in use, and unplug chargers and other devices that draw power when idle.
3. **Heating and Cooling**: Use programmable thermostats to regulate heating and cooling, and properly insulate homes to reduce energy loss.
4. **Use always pressure cooker.**
5. **Grow trees** near the houses and get a cool breeze and shade. This will cut off your electricity charges on AC and coolers.

7. Transportation: Use public transportation, carpool, bike, or walk whenever possible to reduce fuel consumption and emissions.

8. Energy Audits: Conduct energy audits to identify areas where energy can be saved, such as through better insulation or more efficient lighting.

9. Renewable Energy: Consider using renewable energy sources such as solar panels or wind turbines to generate electricity.

10. **Reduce, Reuse, Recycle**: By reducing consumption and reusing items, individuals can indirectly reduce the energy used in the production and disposal of goods.

11. Education and Advocacy: Educate others about the importance of energy conservation and advocate for policies and practices that promote energy efficiency.

12. Lifestyle Choices: Make conscious choices in daily life to reduce energy consumption, such as using natural light during the day and adjusting thermostats to moderate levels.

Thank you